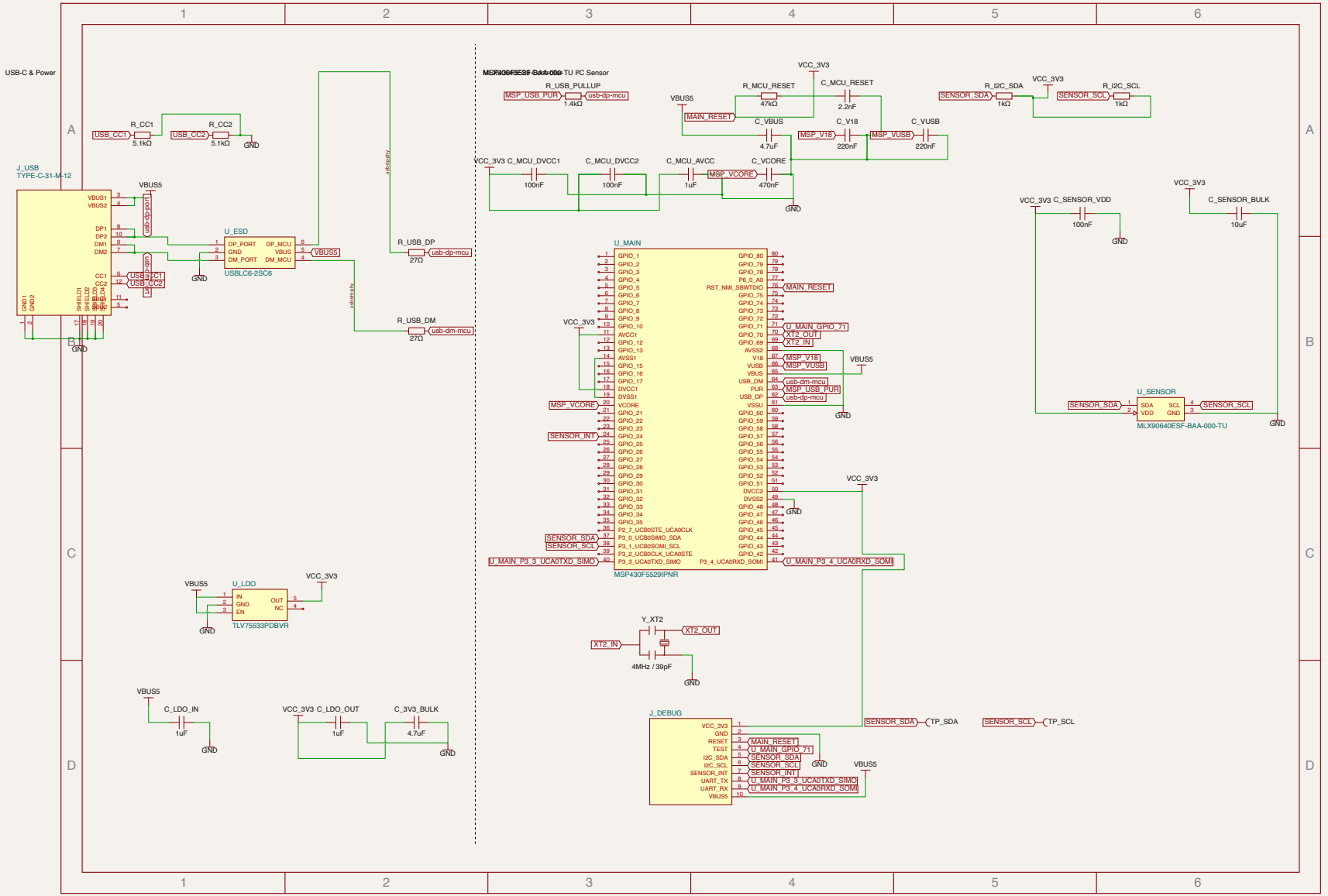
USB-C & Power
MSP430F5529 + MLX90640 I²C Sensor
J_USB
TYPE-C-31-M-12
R_CC1
5.1kΩ
R_CC2
5.1kΩ
USB_CC1
USB_CC2
VBUS5
U_ESD
USBLC6-2SC6
R_USB_DP
27Ω
R_USB_DM
27Ω
usb-dp-pcb
usb-dm-pcb
usb-dp-mcu
usb-dm-mcu
U_LDO
TLV75533PDBVR
VCC_3V3
C_LDO_IN
1uF
C_LDO_OUT
1uF
C_3V3_BULK
4.7uF
R_USB_PULLUP
1.4kΩ
MSP_USB_PUR
R_MCU_RESET
47kΩ
C_MCU_RESET
2.2nF
MAIN_RESET
C_VBUS
4.7uF
C_V18
220nF
MSP_V18
C_VUSB
220nF
MSP_VUSB
C_MCU_DVCC1
100nF
C_MCU_DVCC2
100nF
C_MCU_AVCC
1uF
C_VCORE
470nF
MSP_VCORE
R_I2C_SDA
1kΩ
R_I2C_SCL
1kΩ
SENSOR_SDA
SENSOR_SCL
C_SENSOR_VDD
100nF
C_SENSOR_BULK
10uF
U_SENSOR
MLX90640ESF-BAA-000-TU
U_MAIN
MSP430F5529IPNR
SENSOR_INT
U_MAIN_GPIO_71
XT2_OUT
XT2_IN
U_MAIN_P3_3_UCA0TXD_SIMO
U_MAIN_P3_4_UCA0RXD_SOMI
Y_XT2
4MHz / 39pF
J_DEBUG
TP_SDA
TP_SCL
GND